B.Sc. Engg./HD CSE 4th Semester | 28 October 2019

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | **SUMMER SEMESTER, 2018-2019**
DURATION: 3 HOURS | **FULL MARKS: 200**

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.
There are **8 (eight)** questions. Answer any **6 (six)** of them.
Figures in the right margin indicate marks.

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1. a) What is data communication? Describe the fundamental characteristics of effective data communication system. 2+6

   b) Briefly explain the necessity of layering in designing a communication system. Match the following to one or more layers of the OSI model: 5+5
   i. Topology formation ii. Route selection iii. Framing iv. Addressing v. Flow control

   c) What is multiplexing? Describe the goals of multiplexing. How does multiplexing differ from bandwidth spreading? 2+5+3.33

   d) A nonperiodic composite signal has a bandwidth of 200 kHz with a middle frequency of 140 kHz and peak amplitude of 20 V. The two extreme frequencies have an amplitude of 0. Draw the spectrum in frequency domain. 5

2. a) Write short notes on any **two** of the followings: 4×2
   i. B8ZS ii. Signal-to-Noise Ratio (SNR) iii. MLT-3

   b) Consider a channel having highest bitrate of 12 Mbps and bandwidth of 2 MHz. Calculate the approximate signal level and needed SNR value for this channel. 5

   c) With necessary diagrams and equations, explain the Pulse Code Modulation (PCM) technique for converting an analog signal to a digital signal 10.33

   d) Briefly explain the quadrature amplitude modulation (QAM) technique for digital to analog conversion? Give constellation diagram for the following: 5+5
   i. Binary ASK ii. BPSK iii. QPSK iv. 4-QAM v. 16-QAM

3. a) What do you mean by switching? Compare the delay and efficiency of a datagram network and a Virtual-circuit network. 3+6.33

   b) Mention the major drawback of a crossbar switch? Design a three-stage, 200X200 switch (N = 200) with k = 4 and n = 20 using the Clos criteria. 2+5

   c) Compare and contrast the Stop-and-Wait Protocol with the Stop-and-Wait ARQ protocol. Explain the reason for moving from the Stop-and-Wait ARQ Protocol to the Go-Back-N ARQ Protocol. With necessary example, prove that the send window size for '*Selective Repeat ARQ*' protocol can be at best 2^{m-1}, where m is the size of sequence number. 5+4+8

4. a) Write short notes on any two of the followings: 4×2
   i. Checksum ii. Cyclic Codes iii. Two-dimensional parity check.

   b) What do you mean by minimum Hamming distance? Find the minimum Hamming distance for detection of 6 errors and correction of 2 errors. Detect and correct the single error in the received Hamming codeword **10110010111** (assume even parity). 2+3+6.33

   c) Illustrate the structure of the CRC encoder and the CRC decoder using block diagrams. Given the dataword 1010011010 and the divisor 10111 , 6+4+4

   i. Show the generation of the codeword at the sender site (using polynomials).

   ii. Show the checking of the codeword at the receiver site (using polynomials).

5. a) Explain the co-channel interference and system capacity of a cellular network with appropriate figure and equation. 11.33

b) Briefly explain the concept of Trunking and Grade of Service (GoS). 8

c) A small city of 75000 residents has two competing mobile networks company named A and B that provide cellular service to the users. Company A has 50 cells, each with 40 channels and company B has 100 cells, each with 20 channels. Find the number of users that can be supported at 5% blocking probability if each user averages 2 calls per hour at average call duration of 6 minutes. Compute the percentage market penetration of each company assuming that both the companies are operated at maximum capacity. 8

d) A system has 1000 cells with 25 traffic channels available where a minimum SIR of 15dB must be maintained. Consider that there are 6 channels in the first tier. Find the minimum cluster size with path loss exponent 3. If each user averages two calls per hour at an average call duration of 3 min, how many subscribers can this system support for a 1% GOS ? 6

6. a) Neatly sketch the GSM system architecture and briefly explain the major functional units. 4+6.33

b) Show the process of constructing a GSM frame of 156.25 bits by using channel coding and interleaving. 11

c) Explain the following terms with appropriate examples: 4X3

i. Duplex distance ii. Multipath fading iii. Adaptive equalization iv. Location area

7. a) Give the names of all logical channels available in GSM. Present the tasks of each common control channel (CCCH). 5+6

b) With the aid of necessary diagram explain how a call to a mobile user initiated by a PSTN subscriber is established. Mention the name of different logical channels used in different stages of call establishment. 10

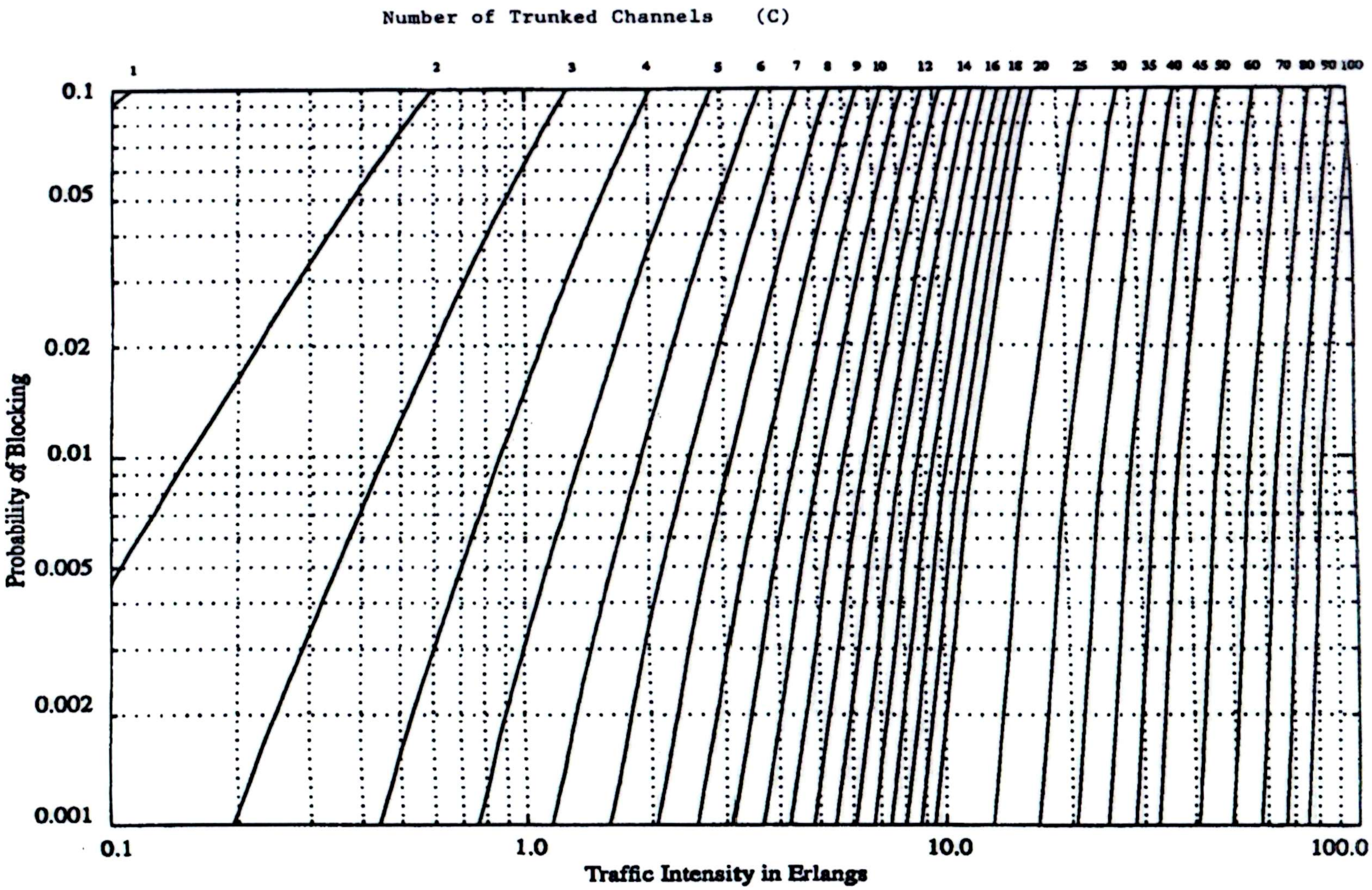
c) With necessary diagrams explain the handoff scenario at a cell boundary. Briefly explain different practical handoff considerations. 6+6.33

8. a) Write short notes on any two of the followings: 5×2

i. Okumura/Hata model ii. Coherence Time iii. Coherence Bandwidth

b) Why do we need propagation models? How does the 'plane- earth loss model' differ from the 'free-space loss model'? How do empirical path loss models differ from deterministic path loss models? 3.33+5+5

c) The speech quality for a mobile communication system is just acceptable, when the received power at the terminals of the mobile receiver is -85 dBm. Find the maximum acceptable propagation loss for the system when the transmit power at the base station is 50 W, base station feeder losses are 5 dB, base station antenna gain is 3 dB. Antenna gain of the mobile is 1 dB and feeder losses at the mobile are 2 dB. Calculate the maximum range of the system using a) the free space loss model, and b) the plane earth loss model, assuming a frequency of 900 MHz and antenna heights of 15 m and 1.5 m. 5+5



The Erlang B chart showing the probability of blocking as functions of the number of channels and traffic intensity in Erlangs.

B.Sc. Engg. CSE 4th Semester 05 April 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **SUMMER SEMESTER, 2020-2021**
DURATION: 3 HOURS **FULL MARKS:200**

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all** of the **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) How are OSI and ISO related to each other? Distinguish between communication at the transport layer, at the network layer, and at the data-link layer. 2+7.33 (CO1) (PO1)

   b) Briefly explain the concept of *Baseline Wandering* and *DC Component* in digital transmission. Find out the bit sequence for the given digital signals from the following figures. For the Figure 1, consider NRZ and NRZ-I coding. And for the Figure 2, consider Manchester and Differential Manchester coding schemes. 6+6 (CO2) (PO1,PO2)

Figure 1 Figure 2

   c) What do you mean by multiplexing? In a synchronous TDM multiplexing technique, different input data rates need to be multiplexed where the output line has a capacity of 200 kbps. Input data rates are 15, 20, 30, 40, 40 kbps. 2+4+6 (CO2) (PO3,PO1)
      i. Draw the block diagram to show how disparity of data rates will be handled.
      ii. Briefly explain each of the techniques.

2. a) Define a switch and describe the need for switching. Compare and contrast a circuit-switched network and a packet-switched network in terms of delay and efficiency. 10.33 (CO3) (PO1,PO2)

   b) What is the role of the address field in a packet traveling through a datagram network? How does the role change in a packet travelling through a virtual-circuit network? 7 (CO3) (PO1,PO2)

   c) Design a three-stage space-division switch with $N = 100$. Use 10 crossbars at the first and third stages and 4 crossbars at the middle stage. (CO3) (PO1,PO3)
      i. Draw the configuration diagram. 4
      ii. Calculate the total number of crosspoints. 2
      iii. Comment on the blocking factor of this system. 2
      iv. State the *Clos* criteria and redesign the above configuration using it. 8

3. a) Write short notes on the followings: 3x3 (CO3) (PO1,PO2)
   i. Checksum ii. Two-dimensional parity iii. Burst error

   b) What do you mean by minimum Hamming distance? Find the minimum Hamming distance for detection of 6 errors and correction of 2 errors. Illustrate the structure of the encoder and decoder for a Hamming code C (7, 4). 3+3+6 (CO3) (PO1,PO2)

c) Define a cyclic code. How does a cyclic code differ from a linear block code? Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using both binary division and polynomials). 2+3+7.33 (CO3) (PO1,PO3)

4. a) What is CDMA? How does CDMA differ from other channelization protocols? Generate the chip sequences for a CDMA network with 14 stations? 2+3+5 (CO3) (PO1,PO3)

b) What is the use of persistence methods in CSMA? Draw the flow diagram of the *p*-Persistent method. 3+4.33 (CO3) (PO1,PO3)

c) Demonstrate a comparative analysis of the random access MAC protocols. (Hints: Your answer must include vulnerable time and throughput along with other parameters as a basis for comparison) 16 (CO3) (PO1,PO2)

OR

With necessary examples and figures demonstrate the evolution process of different link control protocols. (Hints: Your answer should include the window size, acknowledge type along with other parameters) 16 (CO3) (PO1,PO2)

5. a) Define cluster and footprint in cellular communication. With necessary diagrams and equations, explain the frequency reuse concept of cellular communication. 16.33 (CO3) (PO1,PO3)

OR

With necessary diagrams and equations, explain the co-channel interference and system capacity of cellular communication. 16.33 (CO4) (PO1,PO3)

b) A 30 MHz spectrum is allocated to a wireless system that uses two 25 kHz simplex channels to provide full-duplex voice and control channels. Compute the number of channels available per cell if that system uses a 12-cell reuse pattern. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell of that system. 6 (CO4) (PO1,PO2)

c) A system has 800 cells with 25 traffic channels available where a minimum SIR of 15dB must be maintained. Consider that there are 6 channels in the first tier. Find the minimum cluster size with path loss exponent 3. If each user averages two calls per hour at an average call duration of 2 min, how many subscribers can this system support for a 2% GOS? 5+6 (CO4) (PO1,PO2)

6. a) Neatly sketch the GSM system architecture. Name all the logical channels available in GSM and present the tasks of each control channel involved in connectivity maintenance. 4+7 (CO4) (PO1,PO3)

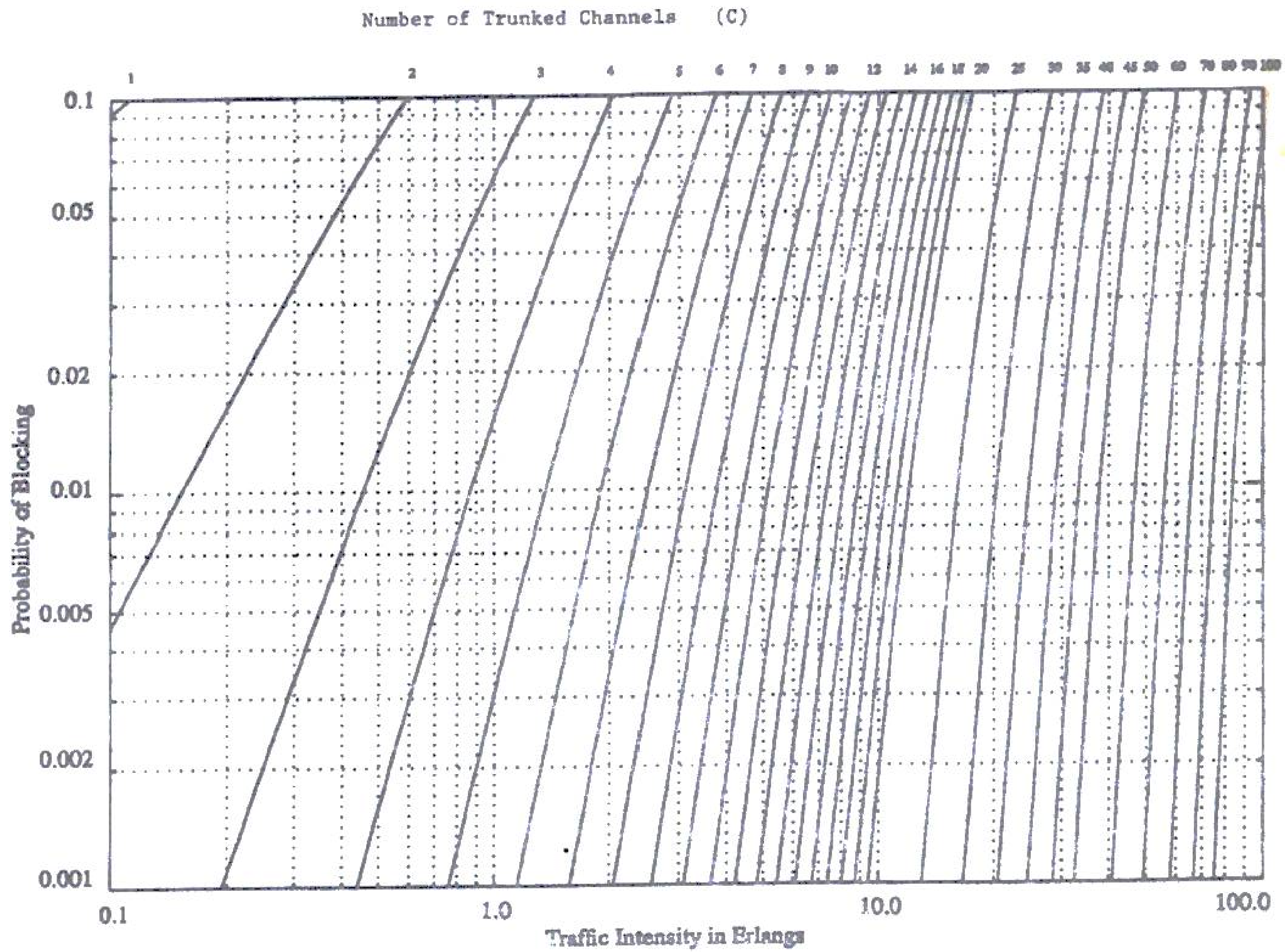
b) With the aid of necessary diagrams, explain how a call to a mobile user initiated by a PSTN subscriber is established. Mention different logical channels involved in different stages of call establishment. 10.33 (CO4) (PO1,PO3)

OR

With necessary diagrams explain the handoff scenario between two cells under different MSCs of a GSM network. Mention different logical channels involved in different stages of hand-off process. 10.33 (CO4) (PO1,PO3)

c) Explain the following GSM terms with appropriate examples: 3×4 (CO4) (PO1,PO3)

i. Interleaving ii. Normal burst iii. Duplex distance iv. Channel coding



The Erlang B chart showing the probability of blocking as functions of the number of channels and traffic intensity in Erlangs.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION | **SUMMER SEMESTER, 2021-2022**
DURATION: 3 HOURS | **FULL MARKS: 200**

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

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1. a) Briefly explain the necessity of layering in designing a communication system. Match the following to one or more layers of the OSI model: 5+
(CC
(PC
i. Transmission mode ii. Flow control iii. Encryption iv. Addressing v. Access control

b) Multiplexing and Spreading are two ways of bandwidth utilization. How do they differ from each other? With necessary diagrams, explain the Frequency Hopping Spread Spectrum (FHSS) technique. How does FHSS differ from Direct Sequence Spread Spectrum (DSSS)? (C
(F

c) Scrambling techniques are used along with line coding schemes to provide synchronization. These techniques insert pulses to break the long sequences of zeros. Briefly explain two common scrambling techniques with the scrambling rules. Consider a bit stream: 1100001000000000 and draw the corresponding digital signals by applying the scrambling techniques. How does block coding differ from scrambling? 13
(C
(F

2. a) Transmission of information in any network involves end-to-end addressing and sometimes local addressing (such as VCI). Table 1 shows the types of networks and the addressing mechanism used in each of them. 4+4+
(C
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Table 1: Type of networks with their addressing mechanism for Question 2.a)

Network	Setup	Data Transfer	Teardown
Circuit-switched	End-to-End		End-to-end
Datagram		End-to-end	
Virtual-circuit	End-to-end	Local	End-to-end

Answer the following questions with necessary examples and diagrams:

i. Why does a circuit-switched network need end-to-end addressing during the setup and teardown phases? Why is no address needed during the data transfer phase for this type of network?
ii. Why does a datagram network need only end-to-end addressing during the data transfer phase, but no addressing during the setup and teardown phases?
iii. Why does a virtual-circuit network need addresses during all three phases?

b) You need to design a three-stage space-division switch with $N = 100$. Use 10 crossbars at the first and third stages and 4 crossbars at the middle stage.

i. Draw the configuration diagram of this system and comment on the blocking factor.
ii. State the Clos criteria and redesign the above configuration using it.

c) Comment on the bandwidth requirements of the following modulation schemes:
i. BFSK ii. BASK iii. Frequency Modulation (FM) iv. Phase Modulation (PM) (CO2) (PO1)

3\. a) Design a half-rate convolution encoder consisting of a shift registrar having three stages and EXOR gates which generate two output bits for each input bit. Draw the state transition diagram and Trellis diagram of the convolution encoder. 12 (CO3) (PO1)

b) Transport Protocol standardized by ISO and CCITT in IS 8072 and X.224 use a powerful method of checksum generation. In this method, two checksum bytes are generated instead of the usual one. The procedure for generating these bytes involves the following steps: 12 (CO3) (PO2)

i. Assume checksum bytes X and Y are 00000000.
ii. Define variables P_i and Q_i such that
$P_i = P_{i-1} + B_i$; where B_i= the i^{th} byte
$Q_i = Q_{i-1} + P_i$; Initial values $P_0 = Q_0 = 0$
iii. P_i and Q_i are calculated for *all the data bytes in the message* and for *initial values of the checksum bytes X and Y*.
iv. From the resulting value of P_i and Q_i variables, calculate the final values of the checksum bytes X and Y as below:
$X = P_l - Q_l$; where l = Number of data bytes +2
$Y = Q_l - 2P_l$

Generate the checksum bytes as per the transport protocol for the following data bytes:

10100101 00100110 11100010 01010101

Regenerate variables P_i and Q_i in the receiving end and show that they are zero if there is no error.

c) What do you mean by minimum Hamming distance? With the aid of block diagrams, illustrate the structure of the encoder and decoder for a Hamming code $C\ (7,4)$. 9.33 (CO3) (PO1)

4\. a) Give the taxonomy of different multiple access techniques. With the aid of a flow diagram, explain the ALOHA protocol. What are the advancements of CSMA protocol over ALOHA protocol? 4+8+5 (CO3) (PO1)

OR

Compare and contrast the 'Stop-and-Wait' protocol with the 'Stop-and-Wait ARQ' protocol. Explain the reason for moving from the 'Stop-and-Wait ARQ' protocol to the 'Go-Back-N ARQ' protocol. With necessary examples, prove that the send window size for 'Selective Repeat ARQ' protocol can be at best 2^{m-1}, where m is the size of sequence number. 5+4+8 (CO3) (PO1)

b) A sender sends a series of packets to the same destination using 'Go-Back-N ARQ'. If the header of the frame allows 5-bit sequence number that starts with 0, what is the sequence number after sending 100 packets? If the sender uses 'Stop-and-Wait ARQ' protocol for flow control then what should be the sequence number after sending 100 packets. 6.33 (CO3) (PO1)

c) What is vulnerable time? Compare the vulnerable time of different random access protocols. 3+7 (CO3) (PO1)

5\. a) What are the rationales behind hexagonal cell geometry for cellular communication? For a hexagonal cell geometry, prove that the co-channel reuse ratio is given by $Q = \frac{R}{D} = \sqrt{3N}$, where the cluster size, $N = i^2 + ij + j^2$. 3.34+10 (CO4) (PO1)

b) Explain the following concepts in terms of cellular communication:
i. Trunking ii. Traffic intensity iii. Cell dragging 3×3 (CO4) (PO1)

c) A system has 800 cells with 25 traffic channels available where a minimum SIR of 15dB must be maintained. Consider that there are 6 channels in the first tier. 5+6 (CO4) (PO1)

i. Find the minimum cluster size with path loss exponent 3.

ii. If each user averages two calls per hour at an average call duration of 2 minutes, how many subscribers can this system support for a 2% GOS?

OR

A small city of 150000 residents has two competing mobile network companies named G and R that provide cellular service to the users. Company G has 50 cells, each with 40 channels and company R has 100 cells, each with 20 channels. Find the number of users that can be supported at 10% blocking probability if each user averages 4 calls per hour at an average call duration of 5 minutes. Compute the percentage market penetration of each company assuming that both the companies are operated at maximum capacity. 11 (CO4) (PO1)

6. a) Neatly sketch the GSM system architecture. Give the taxonomy of GSM logical channels. Briefly explain the authentication process of GSM. 4+4+5 (CO4) (PO1)

b) Draw the normal burst used in GSM. Demonstrate how four GSM bursts (each of 156.25 bits) are constructed from a 20 milliseconds voice signal following the steps of the GSM transmission process. 12.33 (CO4) (PO1)

OR

With the aid of necessary diagram, explain how a call to a mobile user initiated by a PSTN subscriber is established. Mention the name of different logical channels used in different stages of call establishment. 12.33 (CO4) (PO1)

c) Suppose a new mobile communication standard is specified as an alternative to GSM with the following frequency specifications: 8 (CO4) (PO1)

Uplink: 1400-1550 MHz **Downlink:** 1600-1750 MHz

The new standard also specifies that two carrier frequencies would be working at 400 KHz distance for better voice quality. As a telecommunication engineer, calculate the following specification of the new standard.

i. Wavelength ii. Bandwidth iii. Duplex distance iv. Number of radio channels

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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **SUMMER SEMESTER, 2022-2023**
DURATION: 3 HOURS **FULL MARKS: 200**

CSE 4405: Data and Telecommunications

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Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) "In data communications, we commonly use periodic analog signals and non-periodic digital signals" - justify the statement. Explain the concept of a digital signal as a composite analog signal. Name different approaches to transmit a digital signal from one point to another. 12 (CO2) (PO1)

   b) In the NRZ encoding scheme, a positive voltage defines bit 1, and a zero voltage defines bit 0. In NRZ-L, the level of the voltage determines the value of a bit. On the other hand, in NRZ-I, a change or a lack of change in the level of the voltage determines the value of a bit. If there is a change, the bit is 1 and it is 0 otherwise. The ideas of RZ (transition at the middle of a bit) and NRZ-L are combined into the Manchester scheme. Differential Manchester, in contrast, combines the ideas of RZ and NRZ-I. There is always a transition in the middle of a bit, but the bit values are determined at the beginning of the bit. Now, explain the following pitfalls and explain in which scenarios of the input signals (a long string of 0 or a long string of 1), the above-mentioned schemes will face these pitfalls: 10 (CO2) (PO1)

      i. DC Component
      ii. Self-Synchronization
      iii. Baseline Wandering

   c) MLT-3 is a differential coding scheme with more than two transition rules that maps one bit to one signal element (similar to NRZ-I). Explain the MLT-3 scheme with a suitable example and a corresponding state transition diagram. Justify the rationale of the greater complexity (three levels and complex transition rules) at MLT-3. 10 (CO2) (PO1)

   d) Give the taxonomy of digital-to-analog conversion techniques. Which of the techniques are most susceptible to noise? Justify your answer. Briefly explain the bandwidth requirements of different analog-to-analog conversion techniques. 10 (CO2) (PO1)

2. a) Find the minimum Hamming distance for the detection of 6 errors and correction of 2 errors. Illustrate how burst error correction can be performed with the Hamming code. 7 (CO3) (PO2)

   b) Define a cyclic code. How does a cyclic code differ from a linear block code? Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender side (using both binary division and polynomials). 10 (CO3) (PO2)

   c) Design a half-rate convolution encoder consisting of a shift registrar and XOR gates which generate two output bits for each input bit. Represent the state transition diagram and Trellis diagram of the designed encoder. Assume a suitable example and demonstrate how a convolution decoder corrects a single-bit error. 15 (CO3) (PO2)

3. a) We know that both Datagram, and Virtual-circuit need a routing or switching table to find the output port from which the information belonging to a destination should be sent out, but a circuit-switched network does not need such a table. **Give the reason for this difference.** An entry in the switching table of a Virtual-circuit network is normally created during the setup phase and deleted during the teardown phase. In other words, the entries in this type of network reflect the current connections and the activity in the network. In contrast, the entries in a routing table of a Datagram network do not depend on the current connections; they show the configuration of the network and how any packet should be routed to a final destination. The entries may remain the same even if there is no activity in the network. The routing tables, however, are updated if there are changes in the network. **Explain the reason for these two different characteristics.** Compare the packet-delay and efficiency of a Datagram network and a Virtual-circuit network. 10 (CO3) (PO2)

b) With necessary examples and flow diagrams, demonstrate the evolution process of different link control protocols for the noisy channels. Your answer should include the window size, and acknowledgement type along with other parameters. 15 (CO3) (PO2)

c) A sender sends a series of packets to the same destination using the *Go-Back-N ARQ*. If the header of the frame allows a 5-bit sequence number that starts with 0, what is the sequence number after sending 100 packets? If the sender uses the *Stop-and-Wait ARQ* protocol for flow control, then what should be the sequence number after sending 100 packets? 6 (CO3) (PO2)

4. a) What is CDMA? How does CDMA differ from other channelization protocols? Generate the chip sequences for a CDMA network with 17 stations. 10 (CO3) (PO2)

b) Briefly explain the persistent methods used by CSMA protocol. Considering the P-persistent method, draw the flowchart of CSMA/CD protocol. 10 (CO3) (PO2)

c) What is vulnerable time? Explain why the vulnerable time in ALOHA depends on frame transmission time (T_{Fr}) but in CSMA depends on propagation time (T_p). 10 (CO3) (PO2)

5. a) Neatly sketch the GSM system architecture. Suppose, a new mobile communication standard (GSM-3500) is specified as an alternative to GSM-1800 with the following frequency specifications: 10 (CO4) (PO2)

Uplink: 3400-3600 MHz
Downlink: 3800-4000 MHz

The new standard also specifies that two carrier frequencies would be working at 400 KHz distance for a better voice quality. As a telecommunication engineer, calculate the following specifications of the new standard.

i. Wavelength
ii. Bandwidth
iii. Duplex Distance
iv. Number of Radio Channels

b) Give the taxonomy of all logical channels available in GSM. Present the tasks of each of the Common Control Channels (CCCHs). Name the logical channels involved in the handoff process. 10 (CO4) (PO2)

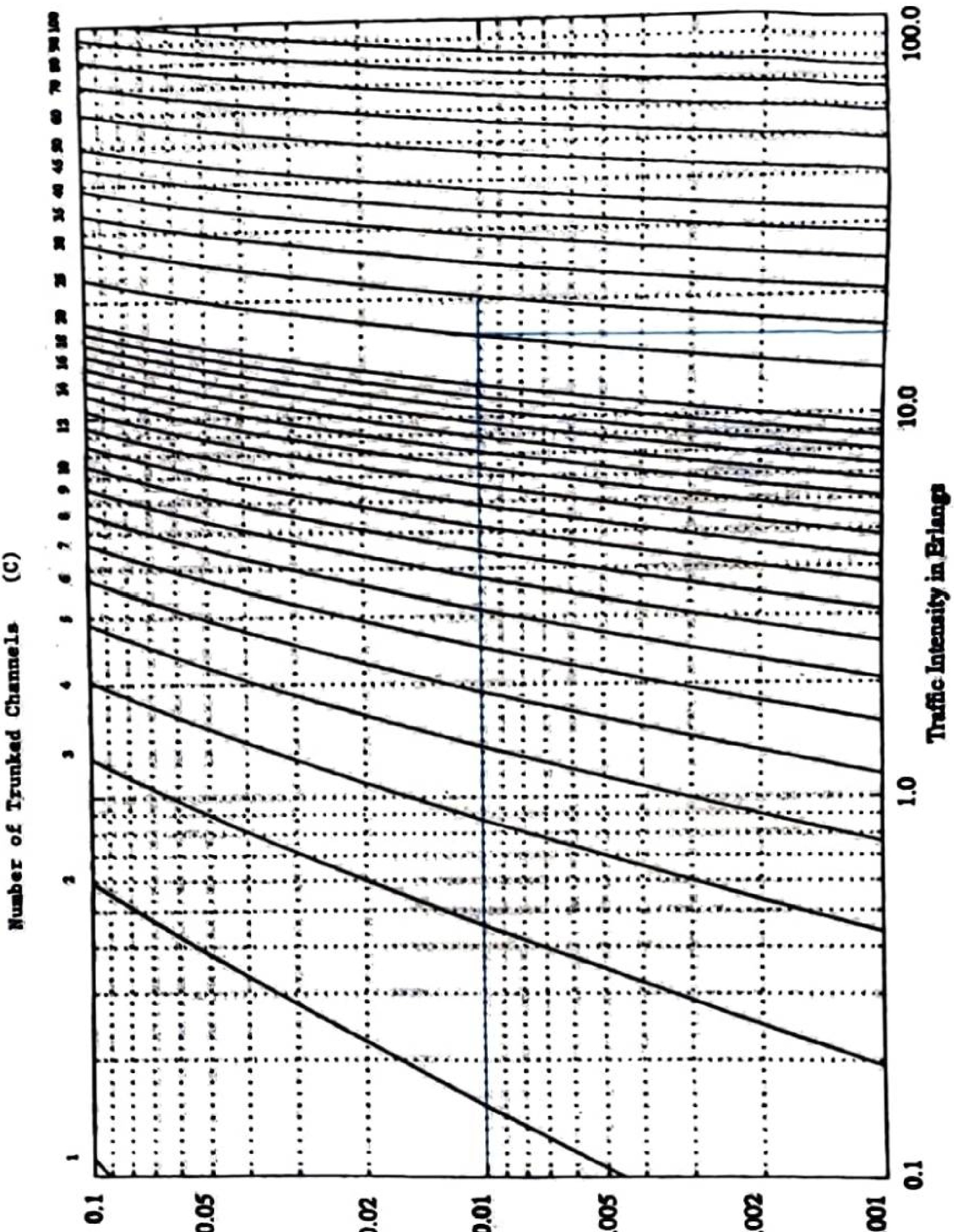
c) Draw a normal burst used in GSM. What is the significance of using the Training sequence (T) in a GSM burst? Demonstrate how four GSM bursts (each of 156.25 bits) are constructed from a 20 milliseconds voice signal following the steps of the GSM transmission process. 15 (CO4) (PO2)

6. a) What are the rationales for hexagonal cell geometry for cellular communications? Explain the co-channel interference and system capacity of a cellular network with appropriate figure and equation. 10 (CO4) (PO2)

b) With necessary diagrams, explain the handoff scenario at a cell boundary. Briefly explain different practical handoff considerations. 10 (CO4) (PO2)

c) Briefly explain the concepts of Trunking Theory and Grade of Service (GoS). A cellular system has 1000 cells with 25 traffic channels available where a minimum SIR of 15dB must be maintained. Consider that there are 6 channels in the first tier. Find the minimum cluster size for a path loss exponent 3. If the average call initiation rate of a user is two calls per hour and the average call duration is 3 minutes, how many subscribers can this system support for a 1% GoS ? 10 (CO4) (PO2)

Appendix



The Erlang B chart showing the probability of blocking as functions of the number of channels and traffic intensity in Erlangs.

B.Sc. Engg. SWE 4th Semester 28 October 2019

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **SUMMER SEMESTER, 2018-2019**
DURATION: 3.0 Hours **FULL MARKS: 150**

CSE 4411: Data Communication and Networking

Programmable calculators are not allowed. Do not write anything on the question paper.
There are 8 (eight) questions. Answer any 6 (six) from the rest of them.
Figures in the right margin indicate marks.

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1. a) Write down the names of protocols working in each layers of TCP/IP protocol suite. 10
   b) Why Slotted AlOHA protocol performs better than the Pure ALOHA protocol? Use necessary diagram to explain. 8
   c) What do you understand by Shannon Capacity? Consider you have a channel with bandwidth 1.0 MHz. The SNR for this channel is 63. What are the appropriate bit rate and signal levels. 7

2. a) Draw the flow diagram of Go-Back-N ARQ. Diagram should contain window state of Sender and Receiver site and timer info. Assume that A is sender and B is receiver. In the following order the events are occurred. Sending window size is four. 10

Table 1: Events for Question 2(a)

1	Initial State	7	Frame 3 is sent but lost
2	Frame 0 is sent, received successfully	8	Timer for Frame 1 expires, Frame 1 resent, received successfully
3	ACK 1 is received successfully	9	Timer for Frame 2 expires, Frame 2 resent, received successfully
4	Frame 1 is sent, received successfully	10	Timer for Frame 3 expires, Frame 3 resent, received successfully
5	ACK 2 is lost		
6	Frame 2 is sent, received successfully but ACK is lost		

   b) What is the window size of Sender and Receiver site in Selective Repeat ARQ? Explain necessary arguments to support your answer. 8
   c) What is bit stuffing and why it is necessary? Assume that the data from upper layer is 7

0011 1110 1111 1111 1100

Show the bit stuffing and un-stuffing process marking the changes.

3. a) Explain the four performance measure of a network: Bandwidth, Throughput, Delay, Bandwidth Delay Product 10
   b) Write short notes on the following in terms of **OSI** model. 9
      i) Process to process delivery
      ii) Host to host delivery
      iii) Source to destination delivery
   c) 'IPv4 is a best effort delivery service'- Explain this statement with appropriate reasoning. 6

4. a) Encode the bit pattern **1011 0011** into following encoding techniques 15
      i. NRZ-I
      ii. Polar RZ
      iii. Manchester
      iv. Differential Manchester
      v. AMI

b) Using $W_1 = [-1]$, Find the chips for a network with **ten** stations. Evaluate with two examples whether the created code follow the characteristics of CDMA. 10

5. a) Discuss three phases of virtual circuit network with appropriate diagrams. 12
   b) What is linear block code? The minimum Hamming distance of a linear block code is the number of 1s in the nonzero valid codeword with the smallest number of 1s. – explain why? 5
   c) In case of Hamming code, we need a dataword of at least 12 bits. Calculate values of *k* and *n* that satisfy this requirement. 4
   d) How to handle a burst error? Explain with example. 4

6. a) Explain the procedure of CSMA/CD protocol. Your answer must contain the Flow diagram. 15
   b) In the Figure 1, the data rate is 10Mbps, the distance between station A and C is 2000m and the propagation speed is 2×10^8 m/s. Station A starts sending along frame at time $t_1=0$, station C starts sending a long frame at time $t_2 = 3\mu s$. The size of the frame is long enough to guarantee the detection of collision by both stations. Find: 6
      i. The time when station C detects the collision (t_3)
      ii. The time when station A detects the collision (t_4)
      iii. The number of bits station A has sent before detecting the collision.
      iv. The number of bits station C has sent before detecting the collision.

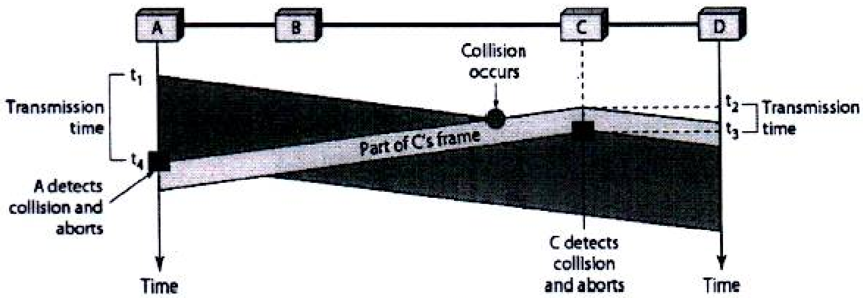

Figure 1: figure for Question 6(b)

   c) Is CSMA/CD is suitable for wireless communication? Why or why not? 4

7. a) Discuss three strategies to maintain compatibility while transitioning from IPv4 system to IPv6 System. 9
   b) What is NAT? In which scenario NAT is necessary? Explain the translation process where both IP address and port address is used. 8
   c) For a class C network address **192.168.10.0** and subnet mask **255.255.255.192** find out the following 8
      i. The Subnet Addresses
      ii. The first valid host for each subnet
      iii. The last valid host for each subnet
      iv. The broadcast address

8. a) Discuss **both** of the TCP connection termination process with appropriate diagrams. 12
   b) 8

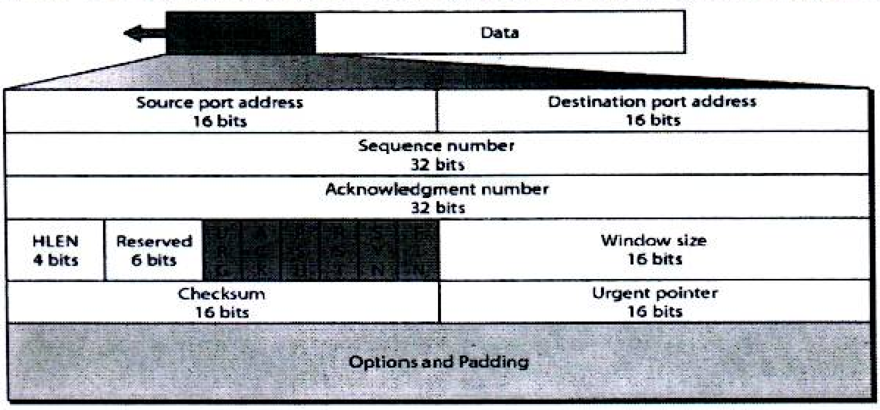

Figure 2: TCP Header Format

The following is a dump of a TCP header in hexadecimal format.

05320017 00000001 00000000 500207FF 00000000

a. What is the source port number?
b. What is the destination port number?
c. What is the sequence number?
d. What is the acknowledgment number?
e. What is the length of the header?
f. What is the type of the segment?
g. What is the window size

c) What is SYN Flooding Attack? How different implementations handle this attack? 5

DTE 2nd Semester 11 April 2022

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **SUMMER SEMESTER, 2020-2021**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4279: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1\. a) In IUT there are more than 2500 internet users. However, IUT has only 4 public IP addresses. Explain the procedure with necessary diagram by which IUT can provide services to 2500 user with only 4 registered public IP. 15 (CO3) (PO3)
What are the advantages an IUT internet user is facing for this?

b) In internet, different network follows one of the two IP protocols - IPv4 or IPv6. Why different protocol is chosen and how can they prevail simultaneously. 10 (CO2) (PO2)

2\. a) An ISP is granted a block of addresses starting with 150.40.0.0/16. The ISP wants to distribute these blocks to 2600 customers as follows: 15 (CO3) (PO3)
- The first group has 200 medium-size businesses; each need 128 addresses.
- The second group has 400 small businesses; each need 16 addresses.
- The third group has 2000 households: each need 4 addresses.

Design the sub-blocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.

b) In IPv4 there is an 8-bit field called service type. Write down the significance of each bit of the field. 10 (CO1) (PO1)

3\. a) Draw the flow diagram of CSMA/CD and CSMA/CA. Criticize the advantages and disadvantages of these two techniques for different transmission medium. 15 (CO3) (PO3)

b) Draw the TCP/IP protocol suite. 10 (CO1) (PO1)
Hint: In your answer you need to mention different protocols in each layer of TCP/IP protocol suite.

4\. a) TCP protocol is a connection-oriented protocol which undergoes several steps in connection establishment. Show the steps of connection establishment with necessary diagram. 15 (CO3) (PO3)
Analyze the disadvantages of this process and how these disadvantages can be overcome.

b) UDP is connectionless transport layer service. Though this service is unreliable, justify its application in different services including Routing Information Protocol (RIP) 10 (CO2) (PO2)

5\. a) In network layer intermediatory router may fragment a received IP datagram. 15 (CO3) (PO3)
   i. When did a router fragment a datagram? Is it possible to avoid fragmentation?
   ii. What are the changes applied on a received datagram before sending out?
      - Case 1: A non-fragmented datagram needs to be fragmented.
      - Case 2: A fragmented datagram (not the last datagram) needs to be fragmented.
      - Case 3: A fragmented datagram (last fragment) needs to be fragmented.

b) Write down the unabbreviated form of following IPv6 address — 10 (CO1) (PO1)

i. 0::0
ii. 0::FFFF:0:0
iii. 582F:1234::2222
iv. 4821::14:22
v. 54EF::A234:2

6. a) Which data link control protocol is represented by the flow diagram in figure 1. Explain the advantages and disadvantages of this protocol from the given scenario. — 12 (CO3) (PO3)

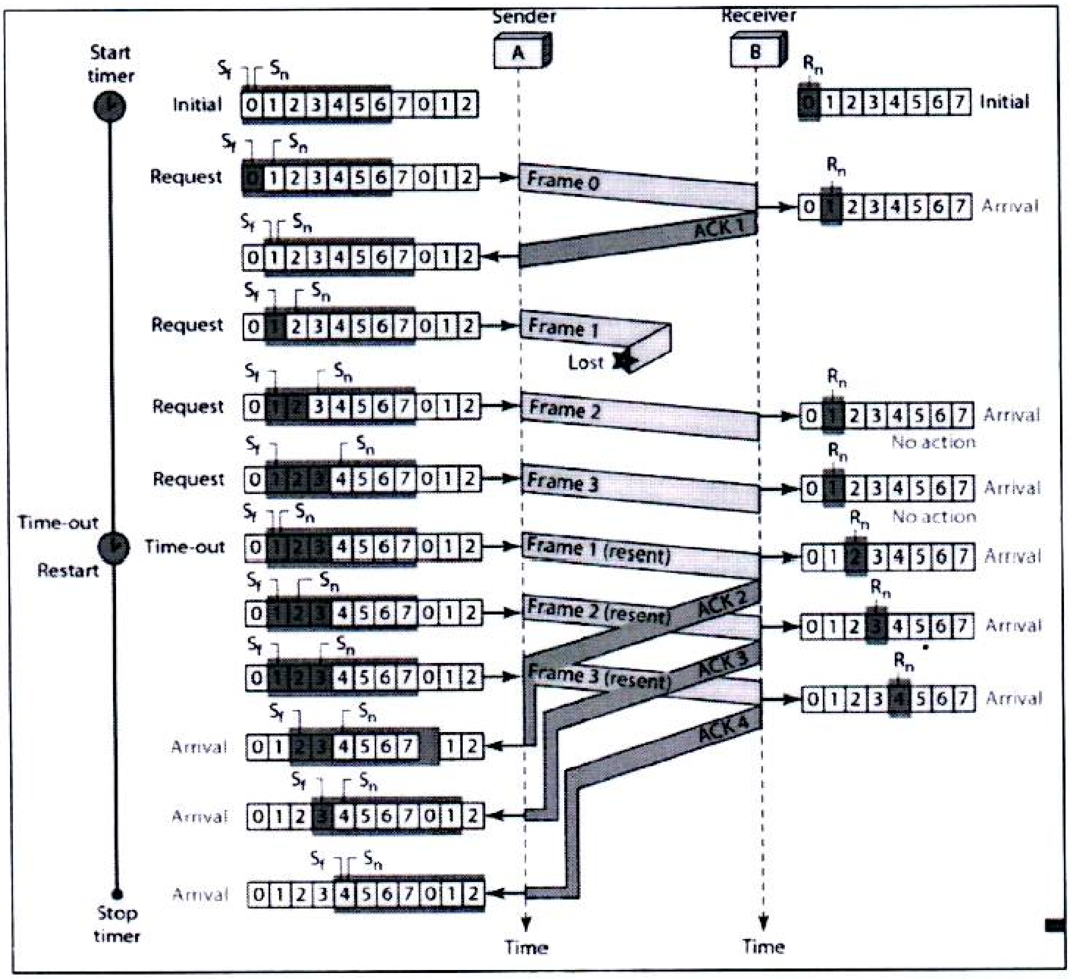

Figure 1: Flow diagram of a data link control technique

b) Explain with necessary diagram how a burst error can be handled with single-bit error correction method. Justify the handling procedure of the burst error with single-bit error correction instead of a unique burst error solution. — 8 (CO2) (PO2)

c) Write down the differences between circuit switched network and packet switched network. — 5 (CO1) (PO1)

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Engg. SWE 4th Semester 16 May 2023

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ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION **SUMMER SEMESTER, 2021-2022**
DURATION: 3 HOURS **FULL MARKS: 150**

CSE 4411: Data Communication and Networking

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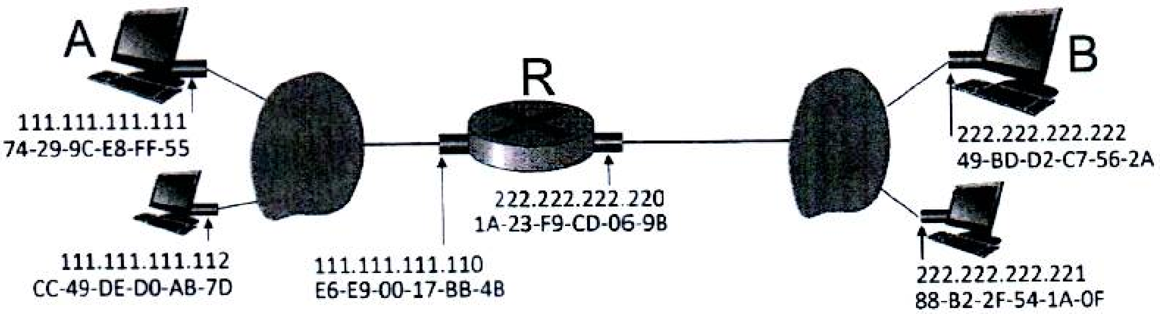

Figure 1: Figure for Question 1. a)

1. a) i. Assume that an ICMP packet is transmitted from end device *A* in one subnet to end-device *B* in another subnet depicted in Figure 1. Write down a walkthrough of the transmission of the packet (i.e, from which hop to which hop the packet will move). What are the source and destination IP address and MAC address present in the frame in each hop? How will the addresses be selected by that hop? Assume that *A* knows *B*'s IP address, IP address of the first hop router *R*, and *R*'s MAC Address. 10+5 (CO2) (PO1)

   ii. How can *A* determine the IP address and MAC address of the first hop router *R* in the scenario mentioned above?

   b) What is the motivation for configuring a VLAN in an institution? Explain with a figure. 10 (CO1) (PO1)

2. a) Explain the protocol of Carrier Sense Multiple Access with Collision Avoidance with the necessary diagram. 10 (CO1) (PO1)

   b) Why is there a constraint on the minimum frame size in CSMA/CD? How is it calculated? Show it with the necessary diagram(s). 5 (CO2) (PO1)

   c) In the network layer, an intermediatory router may fragment a received IP datagram. 4 + 6 (CO2) (PO1)

   i. When does a router fragment a datagram? Is it possible to avoid fragmentation?

   ii. What are the changes applied on a received datagram before sending it out for each of the following cases?
   Case 1: A non-fragmented datagram needs to be fragmented.
   Case 2: A fragmented datagram (not the last datagram) needs to be fragmented.
   Case 3: A fragmented datagram (last fragment) needs to be fragmented.

3. a) Assume that you are using two-dimensional parity code on a 24-bit data word whic
vided into rows of 7-bit. Answer the following questions with one example for each:

   i. What is the maximum number of errors that can be corrected with a guarantee?
   ii. What is the maximum possible number of errors that can be corrected?
   iii. What is the maximum number of errors that can be detected with a guarantee?
   iv. What is the maximum possible number of errors that can be corrected?

b) An IPv4 datagram is under preparation with the following information in the header (in hexadecimal): (CO2)
Ox 45 00 00 54 00 13 58 50 20 06 00 00 7C 4E 03 02 B4 0E 0F 02 (PO1)
See Figure 2 for the format of IPv4 datagram.

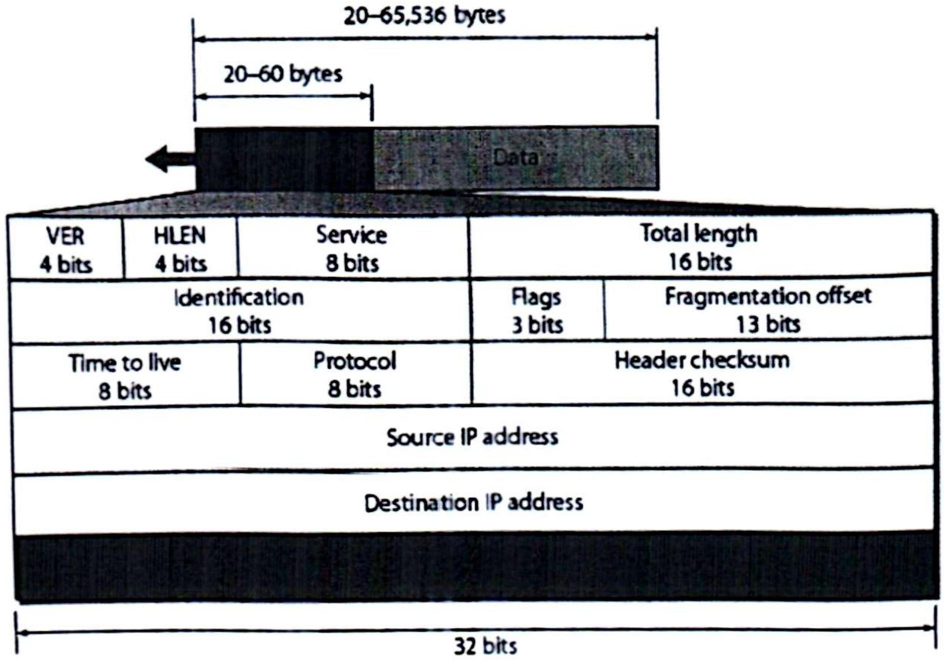

Figure 2: Format of IPv4 datagram for Question 3. b)

The network layer will calculate the header checksum and generate the final IPv4 datagram. Show the steps of header checksum calculation and find out the final IPv4 datagram.

c) How can linear block code can be used to correct burst errors? Explain it with an example. 7
(CO1)
(PO1)

d) Assume that you need to design a hamming code where the dataword needs to be at least 7 bits. Calculate the values of k and n that satisfy this requirement. 3
(CO1
(PO1

4. a) In NRZ encoding scheme, positive voltage defines bit 1 and zero voltage defines bit 0. In NRZ-L, the level of the voltage determines the value of the bit. On the other hand, in NRZ-I, the change or lack of change in the voltage level determines the bit's value. If there is no change in voltage level, the bit is 0. If there is a change, the bit is 1. oNRZ-I scheme is similar to NRZ-I. The only difference is that if there is no change in the voltage level the bit is 1 and if there is a change, the bit is 0. 10
(CO2
(PO1

Now, explain the following pitfalls and scenarios in which these four schemes will face these pitfalls.

- DC Component
- Self-Synchronization
- Baseline Wandering

Scenarios are long strings of 0 and 1 in the input signal.

b) Convert the hexadecimal string **6B 42 27 2F 6A 38** to the 8B6T signal. Draw the signal with notations. You can use multiple lines if it does not fit in a single line. A partial conversion table of 8B6T is given in Figure 3. 10 (CO2) (PO1)

Table F.1 *8B/6T code*

00	-+00-+	**20**	-++-00	**40**	-00+0+	**60**	0++0-0
01	0-+-+0	**21**	+00+--	**41**	0-00++	**61**	+0+-00
02	0-+0-+	**22**	-+0-++	**42**	0-0+0+	**62**	+0+0-0
03	0-++0-	**23**	+-0-++	**43**	0-0++0	**63**	+0+00-
04	-+0+0-	**24**	+-0+00	**44**	-00++0	**64**	0++00-
05	+0--+0	**25**	-+0+00	**45**	00-0++	**65**	++0-00
06	+0-0-+	**26**	+00-00	**46**	00-+0+	**66**	++00-0
07	+0-+0-	**27**	-+++--	**47**	00-++0	**67**	++000-
08	-+00+-	**28**	0++-0-	**48**	00+000	**68**	0++-+-
09	0-++-0	**29**	+0+0--	**49**	++-000	**69**	+0++--
0A	0-+0+-	**2A**	+0+-0-	**4A**	+-+000	**6A**	+0+-+-
0B	0-+-0+	**2B**	+0+--0	**4B**	-++000	**6B**	+0+--+
0C	-+0-0+	**2C**	0++--0	**4C**	0+-000	**6C**	0++--+
0D	+0-+-0	**2D**	++00--	**4D**	+0-000	**6D**	++0+--
0E	+0-0+-	**2E**	++0-0-	**4E**	0-+000	**6E**	++0-+-
0F	+0--0+	**2F**	++0---0	**4F**	-0+000	**6F**	++0--+
10	0--+0+	**30**	+-00-+	**50**	+--+0+	**70**	000++-
11	-0-0++	**31**	0+--+0	**51**	-+-0++	**71**	000+-+
12	-0-+0+	**32**	0+-0-+	**52**	-+-+0+	**72**	000-++
13	-0-++0	**33**	0+-+0-	**53**	-+-++0	**73**	000+00
14	0--++0	**34**	+-0+0-	**54**	+--++0	**74**	000+0-
15	--00++	**35**	-0+-+0	**55**	--+0++	**75**	000+-0
16	--0+0+	**36**	-0+0-+	**56**	--++0+	**76**	000-0+
17	--0++0	**37**	-0++0-	**57**	--++0	**77**	000-+0
18	-+0-+0	**38**	+-00+-	**58**	-+-0++	**78**	+++--0

Figure 3: A partial conversion table of 8B6T for Question 4. b)

c) What is the difference between flow control and congestion control? 5 (CO1) (PO1)

5. a) Each node in the computer network needs a unique IP address. However, it is time-consuming to assign IP addresses to all the nodes manually. Describe the strategy to obtain an IP address automatically by a new client arriving in the network. Your answer must contain the labeled diagram of interactions of different agents participating in the strategy. 12 (CO2) (PO1)

b) Suppose you are interested in detecting the number of hosts behind a NAT. You observe that the IP layer stamps an identification number sequentially on each IP packet. The identification number of the first IP packet generated by a host is a random number, and the identification numbers of the subsequent IP packets are sequentially assigned. Assume all IP packets generated by hosts behind the NAT are sent to the outside world.

i. Assume that you can sniff all the packets sent by the NAT to the outside. Based on the aforementioned scenario, can you outline a simple technique that detects the number of unique hosts behind a NAT? Justify your answer.

ii. If the identification numbers are assigned randomly instead of sequentially, would your technique work? Justify your answer.

6. a) Explain the procedure of a SYN flood attack. How can you prevent this attack? 5 (CO1) (PO1)

b) Figure 4 represents a data transfer protocol. Use this figure to answer the following questions: 4 + 6 (CO2) (PO2)

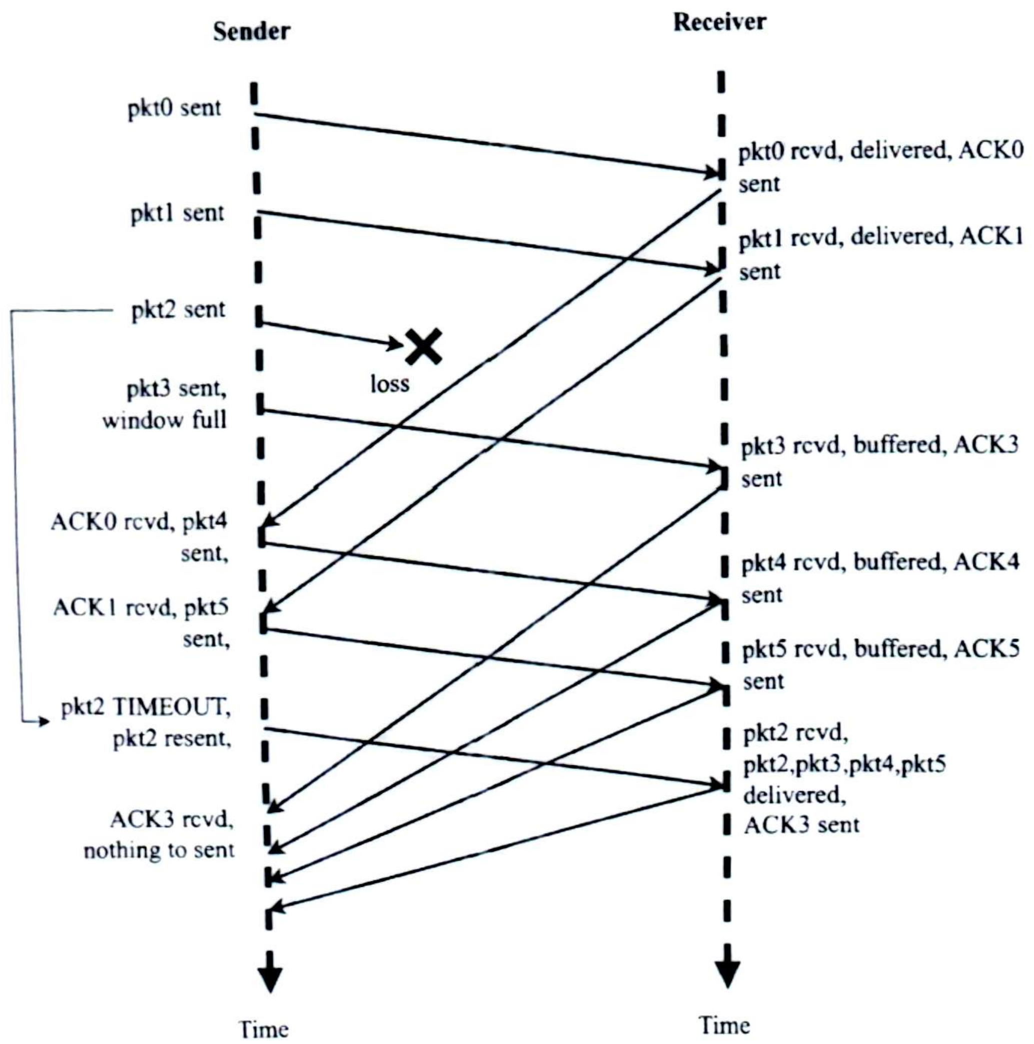

Figure 4: Data transfer protocol for Question 6. b)

i. Which data transfer protocol is represented in Figure 4? Justify your answer.

ii. Redraw Figure 4 with sequence numbers and current window status after each action. Note that you need to choose the sequence number size and window size in such a way that matches the constraints in the figure. You need to draw a box surrounding sequence numbers to represent the current window.

c) Describe the steps of the closing procedure of a TCP connection with a labeled timing diagram of interactions between the client and the server. (CO (PO

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1. John and Tom are communicating using a noisy channel. The protocol being used ensures re-sending of packets in case of failures. However, they observed that communication gets delayed when there are many packets to be sent. Tom identified that this is because if one of the packets gets dropped, all the current packets are resent.

   Now, answer the following questions:

   a) Identify the protocol used in the scenario and suggest an alternative that solves the problem identified by Tom. With necessary diagrams and examples, demonstrate the working procedure of the new approach by sending at least 10 packets. 16 (CO3) (PO3)

   b) Discuss the required sizes of the send and receive windows of the new approach with proper justification. 7 (CO1) (PO1)

   c) Discuss the usefulness of piggybacking in the case of above communication by John and Tom. 7 (CO1) (PO1)

2. Jerry and Simon want to communicate with each other using TCP as the transport layer protocol. There is a server that helps in facilitating the connection. They are aware of the header flags required for connection establishment, connection termination, and data transfer.

   a) Explain the use of `SYN`, `PSH`, `ACK`, and `FIN` flags with proper examples. You can assume appropriate sequence and acknowledgement numbers. 8 (CO1) (PO1)

   b) Why is the UDP still necessary given its unreliability? Provide a real-life scenario where this protocol is used. 6 (CO1) (PO1)

   c) Identify the differences between the TCP and the UDP headers and provide an explanation for the differences. 6 (CO1) (PO1)

3. a) A person is using a lightweight device for accessing a website. Suggest the suitable type of DNS resolution that should be applied for the given scenario. 7 (CO1) (PO1)

   b) A client sends an HTTP request for a document to a server and the server sends the reply. After that, the client sends some information to the server. Identify and briefly describe the methods of HTTP used in this scenario. 9 (CO1) (PO1)

   c) Write short notes on generic, country, and inverse domains. 4 (CO1) (PO1)

4. a) You are the manager of an ISP company. Your ISP is granted a block of addresses starting with 155.55.0.0/16. The ISP wants to efficiently distribute these blocks to 352 customers as follows:

- The first group has 32 customers; each needs 256 addresses
- The second group has 64 customers; each needs 128 addresses
- The third group has 256 customers; each needs 64 addresses

Now, answer the following questions:

i. Design the sub-blocks and show the address allocation and distribution by the ISP. You can ignore the network and broadcast addresses. Find out how many addresses are still available after these allocations. 12 (CO2) (PO2)

ii. Consider that, two customers from the first group need 510 addresses each. What changes can be incorporated to make this possible? 7 (CO1) (PO1)

b) What is the main benefit of using Network Address Translation (NAT)? 6 (CO1) (PO1)

c) Abbreviate the following IPv6 address with proper rules: 5 (CO1) (PO1)

`ABEC:0780:0000:0000:0000:C0EE:0000:AEEA`.

5. a) What are the propagation time and the transmission time for a 3.5 kB message (an e-mail) if the network bandwidth is 0.5 Gbps? Assume that the distance between the sender and the receiver is 15,000 km and light travels at 2.5×10^8 m/s. 6 (CO1) (PO1)

b) A clock has only minute hand and hour hand. Comment on the sampling rate for the clock using the concept of Nyquist Theorem. 7 (CO1) (PO1)

c) Compare the constellation diagrams of 4-QAM and 8-QAM. 5 (CO1) (PO1)

d) You are using an optical fiber cable with wavelengths ranging from 900nm to 1000nm. Each communication requires 10nm of wavelength. How many communications can be accommodated? Discuss the required multiplexing technique. 7 (CO1) (PO1)

6. Renly and Rob want to communicate with each other. As suggested by Rob, they are using a technique where all the physical resources are dedicated for them during their communication.

a) Identify the switching technique they are using. Discuss the feasibility of using this approach if the network is scaled up. 7 (CO2) (PO2)

b) Renly wants to use a technique where allocation of resources are done on demand. Discuss the merits and demerits of this approach. Suggest an alternative approach where both Rob and Renly's ideas can be incorporated together. 6 + 4 (CO1) (PO1)

c) Using the idea of Hamming Distance, find out the size of the codeword if the dataword is of 2 bits and the objective is to detect one or two errors. 8 (CO1) (PO1)